

A Mach-Zehnder Modulator Exceeding 110 GHz in a 300 mm Silicon Photonic Platform towards 400 Gbps/lane Transmission

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Abstract A 420 Gbps PAM-8 Mach-Zehnder modulator with a compact footprint of 500 μm is demonstrated in a 300 mm silicon photonic platform, achieving an insertion loss of 5.6 dB, an electro-optic bandwidth exceeding 110 GHz, and a modulation efficiency of 0.49 V·cm around 1310 nm. ©2026 The Author(s)

Introduction

The rapid proliferation of artificial intelligence (AI), cloud computing, and 5G networks has led to relentless growth in data traffic, driving the need for short-reach (≤ 2 km) optical interconnects with unprecedented capacity and efficiency [1]. This demand is accelerating the transition from 1.6 Tb/s (≈ 200 Gbps/ λ) links to 3.2 Tb/s systems (≈ 400 Gbps/ λ) and beyond. Towards this end, electro-optic (EO) modulators require higher bandwidth, greater modulation efficiency, a more compact size, and lower propagation loss.

Silicon photonics (SiPh) has emerged as a preeminent platform to meet these challenges, leveraging its inherent complementary metal oxide semiconductor (CMOS) compatibility for cost-effective, wafer-scale manufacturing [2]. As a key photonic building block, the silicon modulator performs the essential task of converting electrical signals into optical signals. However, the transmission rate limitations imposed by the finite plasma dispersion effect in silicon modulators have raised concerns about their viability for ultrahigh-speed scenarios, particularly in comparison to novel heterogeneous material platforms, such as thin-film lithium niobate and organic polymers [3].

Furthermore, achieving high integration density presents another challenge for silicon modulators. Mach-Zehnder modulators (Si-MZMs) offer a broad operational wavelength range but are limited in integration density by their millimeter-scale phase shift arms [4]. Micro-ring modulators (Si-MRMs) achieve a compact footprint but their narrow passbands and poor thermal stability require additional feedback mechanisms [5].

Silicon slow-light modulators overcome these limitations by enhancing light-matter interaction through an increased group refractive index [6]. However, conventional slow-light modulators typically exhibit wavelength dependence (due to the slow-light effect's localization at the photonic bandgap edges) or excessive propagation losses (induced by scattering effects) [7-9].

In this paper, a novel high-speed silicon slow-

light modulator operating at telecom wavelengths around 1310 nm is proposed and experimentally demonstrated. Unlike conventional designs based on uniform gratings, this study introduces a unique gradually apodized-grating approach, which significantly reduces mode-mismatch loss and scattering loss. The novel slow-light modulator was realized in a 300 mm silicon photonic platform. Experimental results reveal that the designed slow-light modulators achieve an ultrahigh electro-optic bandwidth (EOBW) exceeding 110 GHz and support 420 Gbps PAM-8 operation (driving voltage: 2.7 V_{pp}). Furthermore, they exhibit a low insertion loss (IL) of 5.6 dB with a compact footprint of 500 μm (a significant improvement over the 10.5 dB reported for a 249 μm device in [3]), alongside a high modulation efficiency of 0.49 V·cm at 1314 nm.

Device Design

The novel slow-light waveguide consists of a series of N_p cascaded apodized-gratings resonators (AGRs), physically equivalent to a coupled-resonator optical waveguide (CROW), as shown in Fig. 1(a). Two adjacent apodized-gratings are separated by a π -phase shift to form the two arms of an AGR, and each arm of the resonator is composed of N_r unit cells (Fig. 1(c)) connected in series. The apodized-grating corrugation depth W_{apo} is defined by the following cosine function:

$$W_{apo} = a \times W_2 \times (1 + \cos(b \times \pi \times (N_r - 1) / N_r)) \quad (1)$$

where W_2 is the offset maximum, b is the apodization strength, and N_r is the number of unit cell. For a conventional resonator, the grating corrugation depth has a constant value W_2 . The comparison of the grating corrugation depth is shown in Fig. 1(d). Simulations of the resonator were performed using the commercial software ANSYS FDTD. The lattice constant Λ is tailored to 243 nm to align the slow-light regime with the O-band window, and other parameters are as follows: $W_1 = 380$ nm, $W_2 = 216$ nm, $W_3 = 105$ nm, the apodization strength $b = 0.8$, and the grating

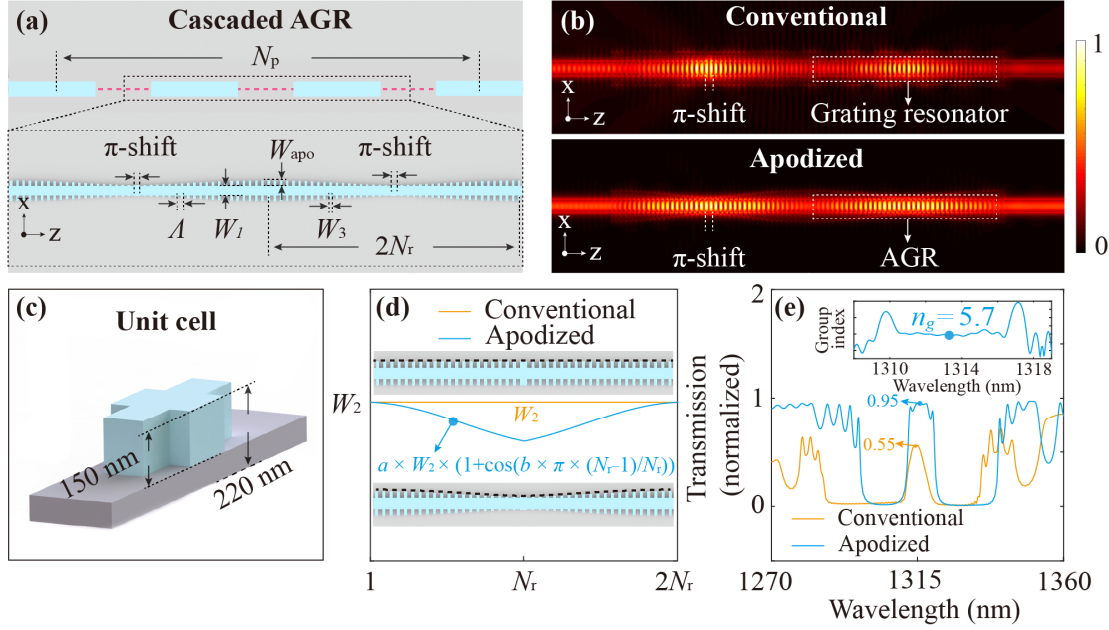


Fig.1: Schematic configuration of the designed slow-light waveguide. a) Top view of the designed novel slow-light waveguide, which consist of a series of coupled apodized-grating resonator. b) Simulated electric field intensity profiles inside the AGR and conventional resonator around 1310 nm. c) Unit cell dimensions of the AGR. d) Comparison of the grating corrugation depth between the conventional resonator and AGR. e) Calculated transmission spectral responses of the conventional resonator and AGR, inset: calculated group index n_g of the AGR around 1310 nm.

unit cell number $N_r = 20$. The normalized transmittance of the slow-light waveguide and the electric field intensity profile for a cascade of $N_p = 5$ resonators are presented in Figs. 1(e) and 1(b) respectively. FDTD simulation results indicate that, compared to a conventional grating resonator, the optical field in the low-loss π -phase shift region of AGR is strongly localized and enhanced, resulting in significantly lower propagation loss. Finally, the group index n_g of the AGR is approximately 5.7 near the wavelength of 1314 nm. Benefiting from lower propagation loss and to balance optical bandwidth and modulation efficiency, the number of resonators N_p in slow-light modulator is set to 50, corresponding to a modulator length of approximately 500 μm .

Device Fabrication and Characteristics

The slow-light modulator was fabricated on a 300 mm SiPh platform. The cross-sectional schematic and an optical micrograph of the modulator are presented in Fig. 2(a). A three-level gradient doping profile tailored for the SiPh platform was applied. Customized light doping concentration of lateral p-n junctions embedded in both modulator arms is $6 \times 10^{17} \text{ cm}^{-3}$ (P) and $8 \times 10^{17} \text{ cm}^{-3}$ (N). The intermediate doping region is 380 nm from the p-n junction with doping concentrations of $2.8 \times 10^{18} \text{ cm}^{-3}$. The doping concentration of the P++ and N++ regions used for the ohmic contact with the metal layer is $1.9 \times 10^{20} \text{ cm}^{-3}$. The right inset of Fig. 1(a) shows the simulated plasma dispersion effect under a -3 V bias, with separate distributions

for electron concentration and hole concentration. As shown in Fig. 2(b), the microwave architecture employs a Ground-Signal-Signal-Ground (GSSG) electrode configuration, with two p-n junctions connected in a series push-pull configuration to halve the effective capacitance and thus guarantee the high EOBW. Fig. 2(c) presents the static transmission spectra, showing a measured IL of slow-light modulator of about 5.6 dB. Additionally, the modulator maintains a flat optical window with a passband width of approximately 7 nm around 1314 nm. The half-wave voltage (V_π) measurements of the fabricated slow-light modulator were characterized by applying a triangular wave. Sweeping the voltage from -8 to 8 V yields a V_π of 9.7 V, corresponding to a modulation efficiency ($V_\pi L$) of 0.49 V·cm. To characterize the EOBW of the modulator, a small-signal measurement system was established, as schematically shown in Fig. 2(e). The measured EO S_{21} and S_{11} curves are presented in Fig. 2(f), showing an EOBW exceeding 110 GHz under a -3 V bias. The EO S_{21} curve rolls off by -1.17 dB at 110 GHz.

The slow-light modulator is further evaluated for high-speed data transmission. Fig. 3(a) illustrates the schematic of the large-signal experimental setup. The eye diagrams are all driven with a peak-to-peak voltage (V_{pp}) of 2.7 V. Figs. 3(b) and (c) present the output eye diagrams for 56 Gbaud and 112 Gbaud NRZ modulation formats, achieving extinction ratios (ER) of

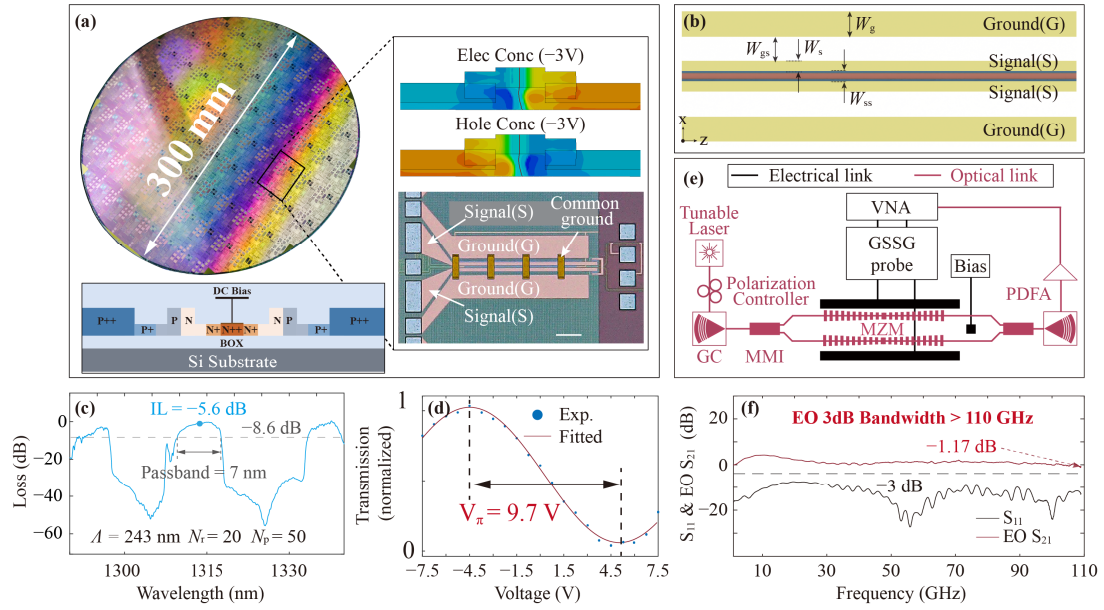


Fig.2: a) Optical photo of the 300 mm SOI wafer and cross-sectional schematic of the modulator. right inset: TCAD numerical simulation of the doping concentration profile in unit cell, along with a micrograph of slow-light MZM. b) Schematic view of the electrode structure. c) Transmission spectrum of the slow-light MZM. d) Normalized optical transmissions of the fabricated slow-light modulator as a function of the applied voltage. e) Experimental equipment for the EOBW measurement. f) The measured S_{11} and EO S_{21} curves.

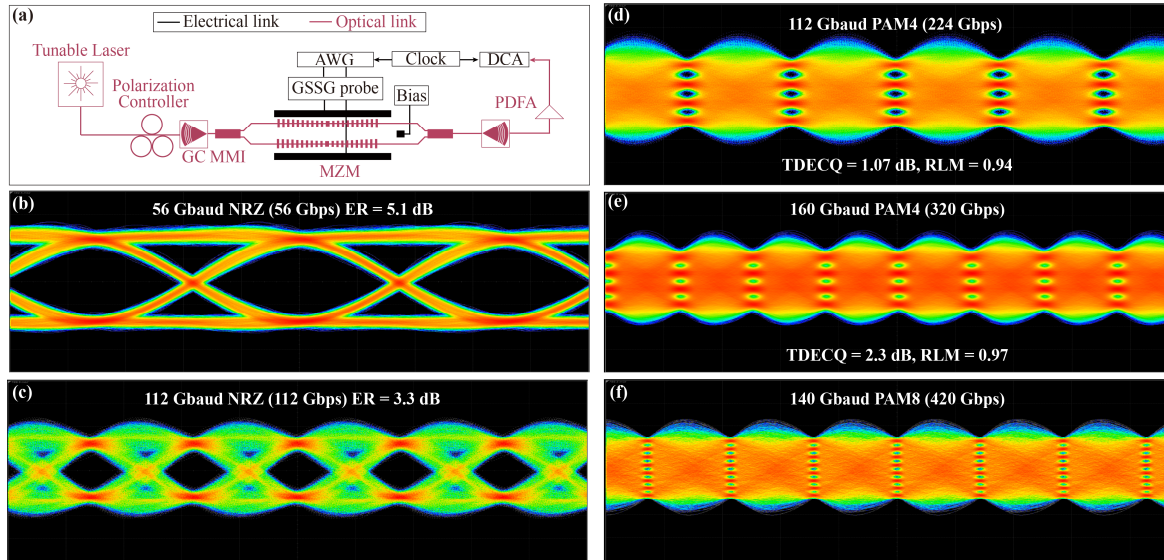


Fig.3: a) Schematic diagram of the experimental setup for the optical transmission experiment. b) and c) The measured optical eye diagram of 56 Gbaud and 112 Gbaud NRZ signals. d) and e) The measured optical eye diagram of 112 Gbaud and 160 Gbaud PAM-4 signals. f) Optical generated 140 Gbaud PAM-8 optical eye diagrams.

5.1 dB and 3.3 dB respectively. Transmitter dispersion eye closure quaternary (TDECQ) measurements of 1.07 dB (5-taps FFE) and 2.3 dB (15-taps FFE) were obtained when driven by 112 and 160 Gbaud four-level pulse amplitude modulation (PAM-4) SSPRQ patterns (Figs. 3(d)-(e)). Fig. 3(f) shows the optical eye diagrams for 140 Gbaud eight-level pulse amplitude modulation (PAM-8), demonstrating a maximum transmission rate of 420 Gbps.

Conclusions

A novel silicon slow-light modulator with a

compact footprint of 500 μm has been demonstrated. The device was fabricated on a 300 mm SiPh platform. By leveraging the slow-light effect in the cascaded AGR optical waveguide, the modulator achieves an IL of only 5.6 dB while maintaining a high modulation efficiency of 0.49 V $\cdot$ cm and an EOBW exceeding 110 GHz. The optical transmission capability of 420 Gbps (140 Gbaud PAM-8) has been experimentally demonstrated. These results further demonstrate the potential of pure silicon modulators for high-speed, high-density interconnects in next-generation networks.

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